Trading Energy Futures with Inflation

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1 Introduction

The following strategies are built using energy futures and various inflation measurements. Since CPI data arrives at low frequency the 5y5y forward inflation measurement works at a good proxy for modeling energy returns. The following futures and time periods will be used.

security	Start Date	End Date
CL1	1990-01-03	2024-10-29
CO1	1990-01-03	2024-10-29
HO1	1990-01-03	2024-10-29
NG1	1990-04-05	2024-10-29
QS1	1990-01-03	2024-10-29
XB1	2005-10-05	2024-10-29

2 Trading Based on β of full Sample OLS

Start by taking the log differences in the inflation measures, lagging them, and regressing them against the returns. Then trading based on the inflation measure scaled by the β

$$r_t = \alpha + \beta \cdot \pi_{t-1,c} + \epsilon \tag{1}$$

Where c is the country. Then take a position by scaling the exogenous signal by the β.

$$\hat{r}_t = \pm \mathbf{1}_{\beta \cdot \pi_{t-1,c}} \cdot r_t \tag{2}$$

Since returns can be volatile and for uniformity and consistency volatility targeting will be used. Due to the high volatile nature of the securities and the low resolution of the data (daily) both perfectly and lagged volatility will be used. For simplicity a 100d exponentially weighting with a 10% volatility targeting. Therefore the resulting strategies are

$$\hat{r}_{p,t} = \pm \mathbf{1}_{\beta \cdot \pi_{t-1,c}} \cdot r_t \cdot \frac{0.1}{\hat{\sigma}_{t,100d}} \tag{3}$$

$$\hat{r}_{l,t} = \pm \mathbf{1}_{\beta \cdot \pi_{t-1,c}} \cdot r_t \cdot \frac{0.1}{\hat{\sigma}_{t-1,100d}} \tag{4}$$

The distinction between the lagged and perfectly volatility targeted returns might seem quite nuanced but it leads to a considerable performance gap. This highlights the underlying nature of energy futures and the need for intraday hedging.

Below are the results of running the regression from eq.1

		Param. Val		p-value		t-stat	
Country	Param. security	α	β	α	β	α	β
UK	CL1	0.000	0.040	0.419	0.178	0.809	1.347
	CO1	0.000	0.033	0.379	0.195	0.880	1.297
	HO1	0.001	0.042	0.098	0.104	1.655	1.628
	NG1	0.000	-0.022	0.656	0.783	0.445	-0.275
	QS1	0.000	0.056	0.287	0.022	1.065	2.286
	XB1	0.001	0.022	0.048	0.453	1.979	0.751
US	CL1	0.000	0.008	0.535	0.640	0.620	0.468
	CO1	0.000	0.016	0.507	0.260	0.663	1.127
	HO1	0.000	0.010	0.203	0.467	1.273	0.728
	NG1	0.000	0.011	0.671	0.808	0.424	0.243
	QS1	0.000	0.011	0.484	0.406	0.699	0.831
	XB1	0.001	0.021	0.055	0.194	1.921	1.299

The p-values and t-stats are not strong, as an attempt to get a better result from the OLS a regression was ran with volatility targeted returns. On the next page are the results of running the regression of the following

$$r_t \cdot \frac{0.1}{\hat{\sigma}_{t,100d}} = \alpha + \beta \cdot \pi_{t-1,c} + \epsilon \tag{5}$$

$$r_t \cdot \frac{0.1}{\hat{\sigma}_{t-1,100d}} = \alpha + \beta \cdot \pi_{t-1,c} + \epsilon \tag{6}$$

Below is table of the regression from eq.5 and eq.6

		Param Val				p-value				t-stat			
	Vol. Target	Lagged		Perfect		Lagged		Perfect		Lagged		Perfect	
	Param	α	β	α	β	α	β	α	β	α	β	α	β
Country	Security												
UK	CL1	0.000	0.009	0.000	0.009	0.712	0.219	0.597	0.223	0.370	1.230	0.529	1.219
	CO1	0.000	0.010	0.000	0.010	0.539	0.168	0.390	0.156	0.614	1.380	0.860	1.418
	HO1	0.000	0.012	0.000	0.011	0.177	0.123	0.067	0.125	1.352	1.543	1.829	1.535
	NG1	0.000	-0.007	-0.000	0.001	0.442	0.712	0.497	0.869	0.769	-0.369	-0.679	0.165
	QS1	0.000	0.015	0.000	0.013	0.553	0.049	0.314	0.051	0.593	1.969	1.007	1.951
	XB1	0.000	0.005	0.000	0.004	0.060	0.527	0.024	0.547	1.881	0.633	2.254	0.603
US	CL1	0.000	0.004	0.000	0.004	0.838	0.294	0.732	0.337	0.205	1.051	0.342	0.961
	CO1	0.000	0.007	0.000	0.006	0.623	0.095	0.487	0.118	0.491	1.668	0.695	1.563
	HO1	0.000	0.005	0.000	0.005	0.304	0.250	0.146	0.205	1.028	1.149	1.454	1.268
	NG1	0.000	0.004	-0.000	0.003	0.455	0.696	0.471	0.506	0.747	0.391	-0.720	0.665
	QS1	0.000	0.007	0.000	0.007	0.546	0.098	0.403	0.078	0.603	1.653	0.836	1.761
	XB1	0.000	0.004	0.000	0.004	0.063	0.303	0.026	0.254	1.857	1.030	2.228	1.141

The p-values and t-stats improve but are still not at an acceptable level. Below on the next page is the returns of trading that each of those strategies.

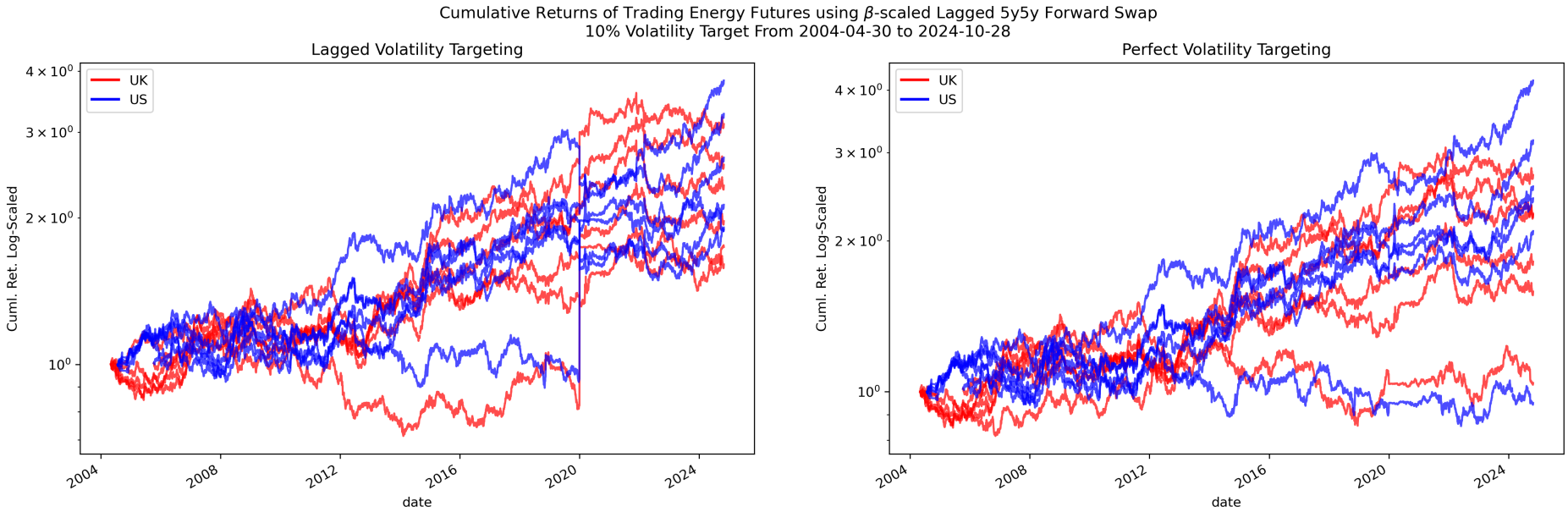

Figure 1: Cumulative returns from β-scaled strategy

The performance of the strategies are reasonable; below is a table of the sharpe. The difference between lagged and perfect volatility targeting highlights how much returns can vary with slightly different volatility hedging methods.

country	group security	Lagged Targetting	Perfect Targetting
UK	CL1	0.289	0.290
	CO1	0.441	0.461
	HO1	0.492	0.474
	NG1	0.201	-0.025
	QS1	0.567	0.553
	XB1	0.413	0.395
US	CL1	0.593	0.557
	CO1	0.657	0.644
	HO1	0.445	0.519
	NG1	0.217	0.037
	QS1	0.665	0.769
	XB1	0.398	0.483

The signal returns do not change the underlying correlation structure of the securities. Below is the table of average cross-correlation between returns controlling for country. Ideally having low correlation across returns is the goal, having high correlation hints that inflation is a heavy driver of returns. In this case neither is present.

group	UK	US	Underlying
Lagged	0.545	0.461	0.482
Perfect	0.460	0.567	0.572

Below is a plot of equal weight across all combined strategies, controlling for volatility targeting and country. Although the returns are equally weighted, the underlying signal returns are volatility targeted. Therefore the portfolio is somewhat volatility targeted.

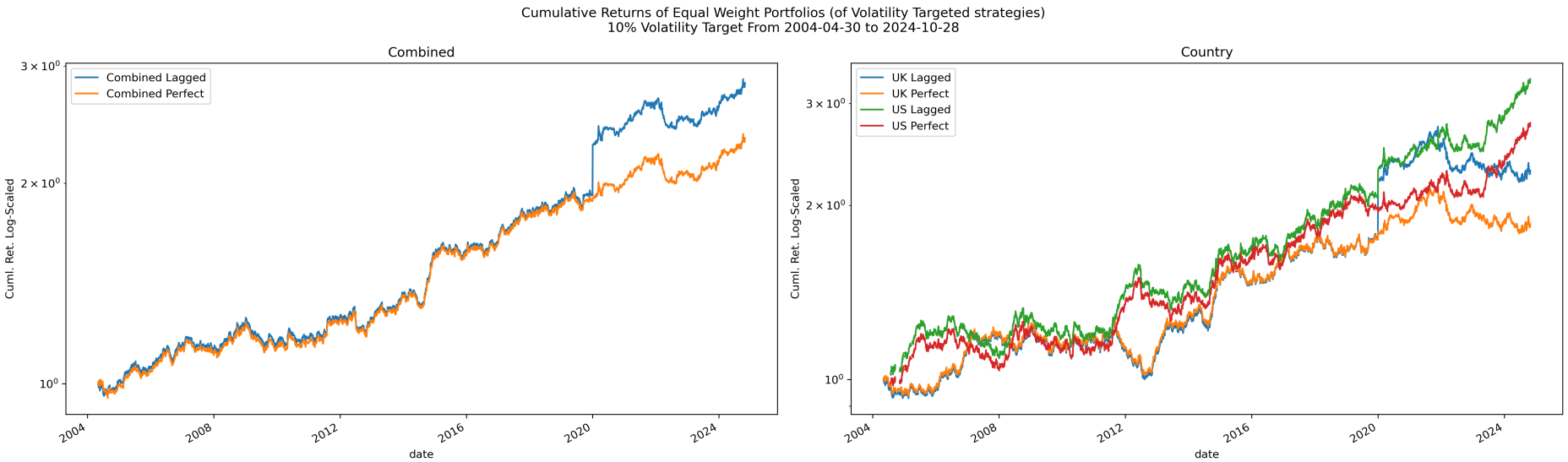

Figure 2: Cumulative returns from combined β

3 Trading Residuals Cross-Sectionally

The next model built is trading a Long-Short portfolio of the same universe except using ranked residuals. Start by running this regression and getting the residuals for security i

$$r_{t,i} = \alpha + \beta \cdot \pi_{t-1,c} + \epsilon \tag{7}$$

$$\epsilon_{t,i} = y_{i,t} - \hat{y}_{i,t} \tag{8}$$

Then rank the residuals, in this case due to the size of the universe 2 buckets will be used a long bucket and a short bucket. Before putting each security in the bucket apply the same volatility targeting method to the underlying securities.

Below is the sharpe of each leg controlling for country and volatility targeting. The results are quite good, since both the long leg has positive sharpe and the short leg has negative sharpe.

Target	Lagged		Perfect	
country	Lower Group	Upper Group	Lower Group	Upper Group
UK	0.741	-0.485	0.896	-0.483
US	0.732	-0.485	0.887	-0.482

On the next page is the performance of each leg while controlling for volatility targeting method and country. The returns across country controlling for leg are relatively the same.

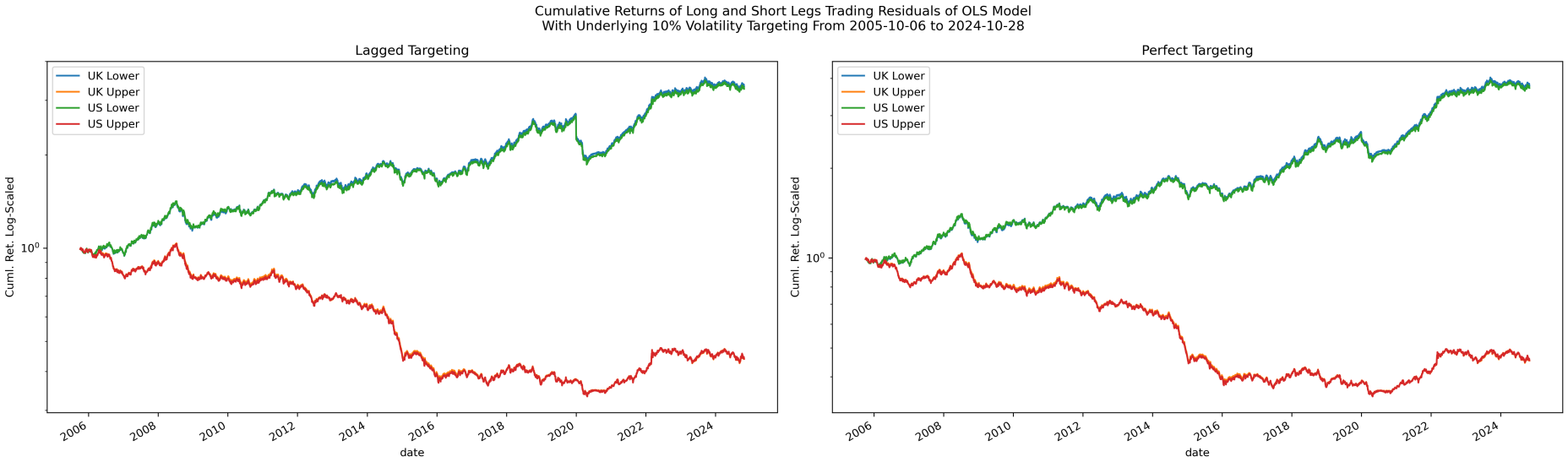

Figure 3: Cumulative returns of each leg of the cross-sectional portfolios

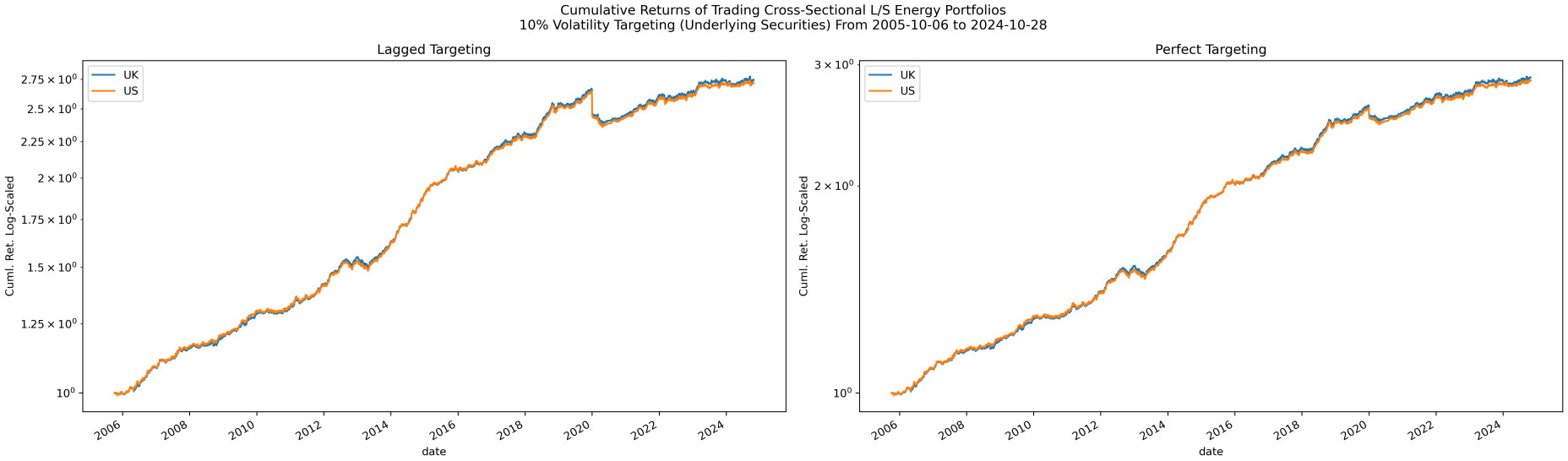

Figure 4: Cumulative returns of trading the long-short portfolio

The sharpes of the long short portfolios are quite substantial and outperform the sharpe of each leg. Below is the table of sharpes.

country	Lagged	Perfect
UK	1.665	2.071
US	1.653	2.054